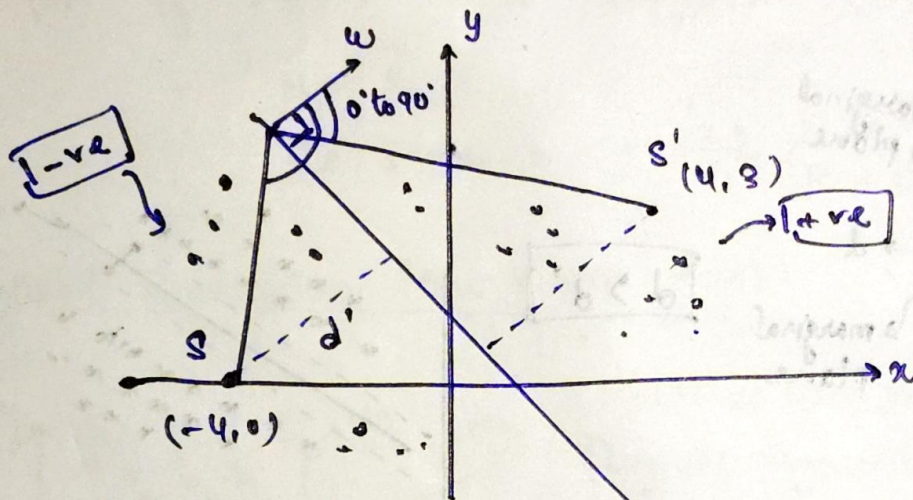


* Support Vector Machine (SVC)



$d = -ve$ below plane
 $d = +ve$ above plane

$$w^T x + b = 0$$

Math Intuition

$$ax + by + c = 0$$

$\Downarrow$

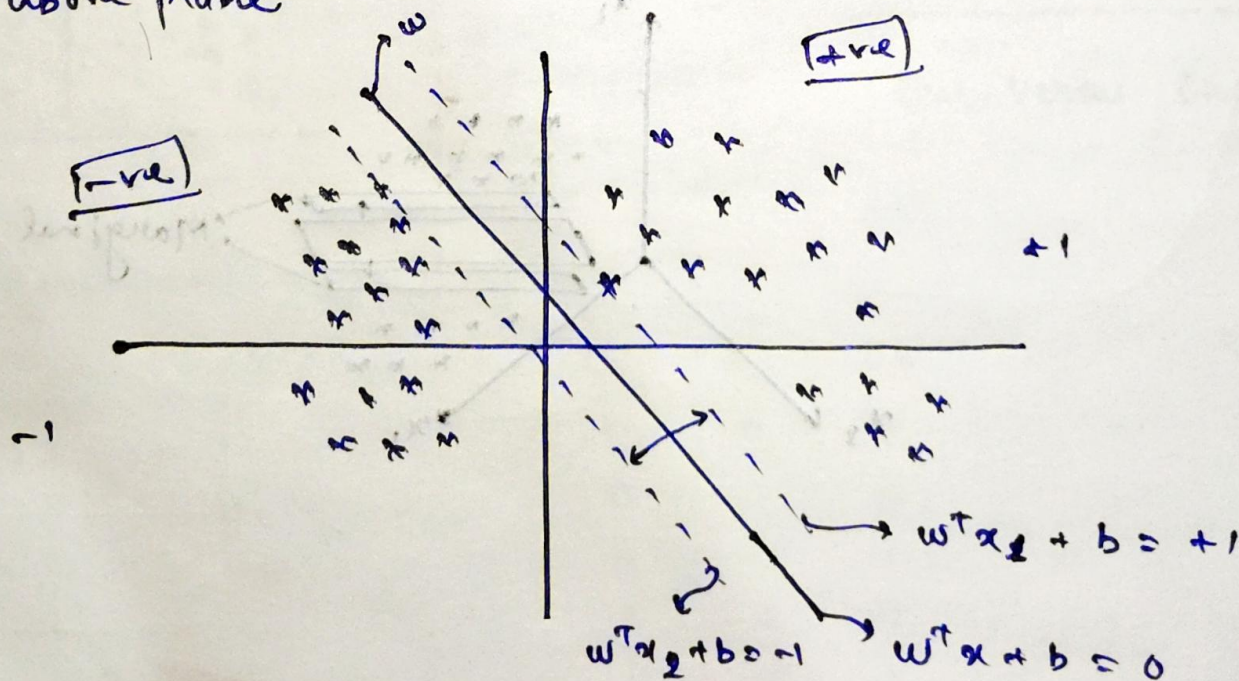
$$w_1 x_1 + w_2 x_2 + b = 0$$

$$w^T x + b = 0$$

, when $b > 0$

$$w^T x = 0$$

→ passes through origin.



$$\begin{aligned} w^T x_1 + b &= 1 \\ (-) \quad w^T x_2 + b &= -1 \quad (+) \\ \hline \frac{w^T (x_1 - x_2)}{\|w\|} &= \frac{+2}{\|w\|} \end{aligned}$$

Unit vector { magnitude of the vector is 1 }

cost function

Maximize $\frac{2}{\|w\|}$ w.r.t $w, b \Rightarrow$ Distance between marginal plane

$\Rightarrow$ correctly classified points

constraint such that

$$y_i \begin{cases} +1 & w^T x + b \geq 1 \\ -1 & w^T x + b \leq -1 \end{cases}$$

For all correct points

constraints $\rightarrow \boxed{y_i (w^T x + b) \geq 1}$